Understanding Public Perceptions and Behaviors Towards COVID-19 Vaccination: A Multifaceted Analysis

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Abstract. The global rollout of COVID-19 vaccines has marked a critical milestone in combating the pandemic. Despite widespread availability, vaccine hesitancy and misinformation continue to challenge public health efforts. Understanding public sentiment towards COVID-19 vaccination is vital to designing effective communication strategies that promote vaccine acceptance. Previous research on vaccine sentiment has largely focused on static snapshots or limited datasets, often neglecting the dynamic and multifaceted nature of public opinions across demographics and regions. Moreover, many studies employ basic classification techniques without exploring advanced feature selection or balancing methods to optimize model performance. This paper aims to bridge these gaps by conducting a comprehensive sentiment analysis on a large-scale Twitter dataset, "COVID-19 Vaccine Tweets," containing over 375,000 unique entries. We apply robust data preprocessing and feature selection methods, including Variance Threshold and SelectKBest, to optimize input features for classification. Using logistic regression as a baseline classifier, we examine sentiment polarity (positive, neutral, negative) and explore the influence of follower count and geographic distribution on public perception. Our study provides deeper insights into evolving vaccine-related discourse and lays the groundwork for more targeted, data-driven public health interventions.

Keywords: Sentiment Analysis; Twitter Sentiment Trends; COVID-19 Vaccination; Natural Language Processing; Public Health Communication.

1. Introduction

Sentiment analysis, a fundamental task in Natural Language Processing (NLP), involves computationally identifying and categorizing opinions expressed in textual data, typically as positive, negative, or neutral. With the exponential growth of digital content on social media, forums, and review platforms, sentiment analysis has become an essential tool for extracting meaningful insights from vast amounts of unstructured text data. This technique enables researchers and

organizations to understand public opinion, predict trends, and make informed decisions based on emotional and attitudinal cues [1, 2].

In the healthcare sector, sentiment analysis offers valuable insights into public attitudes towards health issues, enabling better communication strategies, timely misinformation detection, and enhanced public health interventions. For example, it can be used to monitor perceptions during disease outbreaks, vaccination drives, and policy implementations, providing real-time feedback to health authorities on public trust and concerns [3–5].

While COVID-19 is no longer an immediate global crisis, the methods developed for sentiment analysis during the pandemic remain crucial for managing ongoing and future public health challenges, including vaccination efforts for seasonal diseases, emerging variants, and new health policies. Thus, dynamic sentiment monitoring continues to be a vital component of public health preparedness [6, 7].

However, existing research often limits itself to static or simplistic sentiment analysis, focusing primarily on overall sentiment without adequately capturing how sentiment evolves over time or varies by demographic and geographic factors. Moreover, many studies utilize smaller or imbalanced datasets and lack comprehensive feature optimization, which reduces the reliability and depth of insights.

This study addresses these gaps by conducting a multifaceted analysis of COVID-19 vaccine-related sentiments on Twitter using a large, balanced dataset of over 375,000 unique posts. We apply advanced data preprocessing and feature selection techniques to optimize the dataset and employ logistic regression as a baseline classification method. Importantly, we analyse sentiment dynamics over time and across user characteristics such as follower count and geographic location to provide a richer understanding of public opinion trends.

Our approach leverages established NLP tools, including the VADER sentiment analyser, combined with time-series analysis to track how public sentiment fluctuates in response to key events such as vaccine rollouts and safety announcements. This dynamic, multidimensional perspective enhances the ability of public health authorities to adapt communication strategies promptly, counter misinformation, and foster greater community engagement.

The contributions of this work include:

Utilizing a large, balanced, and well-optimized dataset for improved classification accuracy, Analysing temporal sentiment trends to reveal how public opinion evolves during vaccination campaigns, Investigating demographic and geographic variations in sentiment to identify targeted intervention points, Demonstrating the practical application of sentiment analysis for guiding public health communication strategies. The remainder of the paper is structured as follows: Section 2 reviews relevant literature on sentiment analysis in public health and COVID-19 vaccine research. Section 3 describes the dataset, preprocessing, feature selection, and classification methodology. Section 4 presents results, including tweet activity patterns, sentiment distributions, and dynamic trend analysis. Section 5 concludes the study and suggests directions for future research.

2. Literature Review

The rapid increase in user-generated content on digital platforms has driven the need for advanced Natural Language Processing (NLP) techniques, particularly sentiment analysis, to understand public opinion and social behaviour. Sentiment analysis, which classifies textual data into positive, negative, or neutral categories, plays a crucial role in interpreting societal responses to various events, including public health crises like the COVID-19 pandemic. This section reviews key studies that have applied sentiment analysis to enhance public health communication.

Md. Nawaz Ali et al. (2021) conducted a spatiotemporal sentiment analysis of COVID-19 vaccine perceptions across the United States, revealing persistent negative sentiments in hesitant populations despite an overall trend toward acceptance. By correlating sentiment data with vaccination rates from the CDC and Household Pulse Survey, they identified significant regional variations, especially in less populous states, offering targeted recommendations to overcome vaccination barriers [8].

Bansal et al. (2021) examined public sentiment towards Covaxin and Covishield vaccines during India's second pandemic wave using transformer-based sentiment analysis and word clouds. Their findings captured dynamic shifts in public trust and concerns, emphasizing how health events influence vaccine discourse on social media, especially during outbreaks like the Delta variant [9]. Cheng et al. (2023) analysed Twitter data from England to explore demographic and geographic sentiment variations. They found that while males contributed more positive tweets overall, females over 40 expressed higher positivity. Moreover, a positive correlation between vaccination uptake and tweet sentiment underscored the value of sentiment analysis in informing targeted public health efforts [10].

Molenaar et al. (2024) applied sentiment analysis and topic modelling on Australian tweets related to food security during COVID-19 lockdowns and bushfires. They demonstrated how external events influence public discourse and engagement, highlighting NLP's role in guiding evidence-based policy decisions beyond health crises [11]. Thakur et al. (2024) investigated COVID-19 misinformation on YouTube through sentiment and topic modelling analyses. Identifying themes such as conspiracy theories and political discourse, they revealed content features that drive misinformation spread, illustrating how NLP can support understanding and combating misinformation during health emergencies [12].

Alshanik et al. (2025) conducted a large-scale sentiment and topic analysis of over one million Arabic tweets related to COVID-19. Their work identified key emotional responses and public concerns, such as anger and anticipation, emphasizing the importance of NLP in tailoring effective health communication across different languages and regions [13]. Parveen et al. (2025) studied nearly 870,000 tweets from Ireland and the UK, analysing the relationship between sentiment, message type, user source, and public engagement. Their results showed that personal stories and positive messages generate higher interaction rates, offering insights for crafting impactful public health messages on platforms like X (formerly Twitter) [14].

Collectively, these studies illustrate the power of sentiment analysis and NLP in revealing public emotions, concerns, and behaviours during health crises. They

demonstrate how large-scale social media data analysis can inform targeted and timely public health communication strategies. However, many focus on static sentiment snapshots or narrow aspects of public opinion, highlighting a need for more dynamic, multifaceted analyses. Our study addresses this gap by employing a large, balanced dataset and integrating temporal, demographic, and geographic sentiment analyses to capture evolving public attitudes toward COVID-19 vaccination. This approach aims to provide actionable insights that enhance public health communication and intervention strategies in real time.

3. Methodology

3.1 Dataset Description

The initial phase of this research focused on acquiring a high-quality dataset suitable for comprehensive sentiment and behavioral analysis regarding public discussions about the COVID-19 vaccine. After reviewing multiple sources, we selected a publicly available dataset from Kaggle [15], titled “COVID-19 Vaccine Tweets”. This dataset contains over 100,000 Twitter records across 13 fields, including tweet text, hashtags, location, tweet source, user description, verification status, and timestamp. It specifically comprises tweets tagged with the hashtag #CovidVaccine, capturing diverse public sentiment and discussion during a critical period of the global pandemic. This dataset provides a rich and timely foundation for analyzing social media discourse around vaccination efforts.

3.2 Data Cleaning

Our exploratory analysis identified missing values in key fields: location (20.3%) and hashtags (15.1%). To address these issues: For the location field, which was essential for geographic sentiment analysis, rows with missing location data were removed (exclusion approach). For the hashtags field, missing values were imputed with a placeholder tag, "no_hashtag", to retain the tweets while marking the absence of hashtags. After cleaning, the dataset consisted of 375,155 unique entries. Duplicate tweets, retweets, and irrelevant columns (such as tweet ID and user ID) were removed to ensure high data quality. Further preprocessing of tweet text involved converting text to lowercase, stripping HTML tags, removing URLs, mentions, and special characters using regular expressions (regex). Tokenization, stopword removal, and stemming were conducted using the NLTK library to standardize the corpus and prepare it for advanced analysis.

3.3 Text Processing

Text processing is crucial for transforming raw textual data into a structured analyzable format. The following key steps were applied.

Lowercasing: Converted all text to lowercase to ensure uniformity, as case distinctions are generally irrelevant in sentiment analysis. HTML Stripping: Removed any HTML tags to eliminate noise. URLs and Mentions Removal: Eliminated URLs and user

mentions to avoid irrelevant content. Special Characters Removal: Removed emojis, punctuation, and other special characters. Tokenization: Split text into individual words (tokens) as the basic units for analysis. Stopword Removal: Removed common words such as "the," "is," and "in," which carry little semantic weight for sentiment analysis. Stemming: Applied stemming to reduce words to their root form (e.g., “running” to “run”), reducing data dimensionality while preserving meaning. These preprocessing steps prepared the dataset for sentiment analysis and topic modeling by ensuring data cleanliness and consistency.

3.4 Sentiment Analysis Approach.

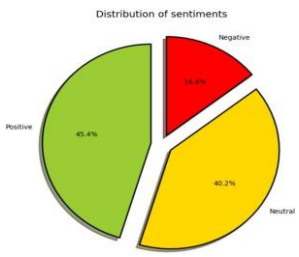
For sentiment classification, we employed the TextBlob library, which uses a lexicon-based scoring system assigning two key metrics per text: Polarity: Ranges from -1 (negative) to +1 (positive), Subjectivity: Ranges from 0 (objective) to 1 (subjective). Based on polarity, tweets were categorized into: Positive: Polarity > 0, Neutral: Polarity = 0, Negative: Polarity < 0.

TextBlob is well-suited for large, informal datasets such as tweets, offering a scalable and fast approach for high-level sentiment insights, though it lacks deeper contextual understanding (e.g., sarcasm detection) [16].

Classification Models and Parameter Settings:

In addition to the lexicon-based method, two supervised machine learning models were used for sentiment classification to compare rule-based and learning-based approaches: Logistic Regression: A widely-used model effective for both binary and multiclass classification, especially with linearly separable data. It outputs probabilities for class membership and serves as a strong baseline. Multinomial Naïve Bayes: Particularly effective for text classification due to its independence assumption and performance with high-dimensional word-frequency data.

Preprocessing and Feature Extraction:
Before training, tweets underwent: Lowercasing and cleaning (removal of URLs, punctuation, stopwords), Tokenization and vectorization using TF-IDF (Term Frequency-Inverse Document Frequency). Parameter Settings: Logistic Regression’s regularization strength was tuned via the C parameter to prevent overfitting, Multinomial Naïve Bayes employed the alpha smoothing parameter to handle zero-frequency features.



Both models were evaluated using cross-validation to ensure robustness and generalizability.

Figure 1 illustrates the sentiment distribution based on TextBlob analysis:

Positive tweets: 45.4%, Neutral tweets: 40.2%, Negative tweets: 14.4%

This distribution indicates a predominance of positive and neutral sentiments, with a relatively smaller negative sentiment proportion. Given the class imbalance, particularly the underrepresentation of negative tweets, we applied SMOTE (Synthetic Minority Oversampling Technique) to generate synthetic examples for the minority class. This balancing improved machine learning model performance by enabling better detection of less frequent sentiment categories.

4. Results and Discussions

4.1 Data Analysis

This section presents a detailed exploration of the dataset across various dimensions.

4.1.1 Tweet Activity by Hour

Examining tweet activity by hour reveals distinct patterns of user engagement throughout the day. A bar chart illustrating tweet frequency over a 24-hour cycle shows significant peaks at 9 AM and 3 PM, indicating increased user activity during these times. This pattern suggests that engagement is highest during standard workday hours, roughly from 9 AM to 5 PM. Understanding these temporal trends is valuable for optimizing the timing of content release to maximize reach and user interaction.

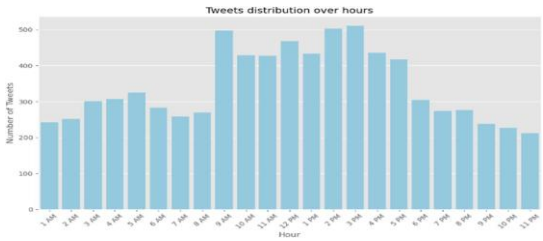


Figure 2. Tweet volume by hour.

4.1.2 Geo-Temporal Analysis

The geo-temporal analysis investigates how sentiment scores vary across different geographic locations, aiming to reveal regional differences in public opinion surrounding vaccines and related health topics. By filtering the dataset to focus on selected countries, a bar plot was generated to visualize sentiment scores, providing insights into how geography influences public attitudes on a global scale [19].

The visualization indicates that countries such as Bulgaria (-0.14), Grenada (-0.13), and Yemen (-0.11) exhibit predominantly negative sentiments, suggesting greater scepticism or criticism toward vaccines and related health issues in these regions. Negative sentiment scores in these countries may reflect challenges such as limited vaccine access, the spread of misinformation, or lower public trust in healthcare systems. Conversely, countries like Bangladesh (0.19), Costa Rica (0.18), and Haiti (0.18) display positive sentiment, indicating more optimistic or supportive attitudes toward vaccination efforts. The positive sentiment in Bangladesh might be attributed to effective public health campaigns or government initiatives, while favourable attitudes in Costa Rica and Haiti could signal successful vaccination drives and high public confidence in health measures.

This geo-temporal sentiment analysis highlights significant regional variations, offering valuable insights that can guide targeted vaccine promotion strategies and tailored public health outreach programs.

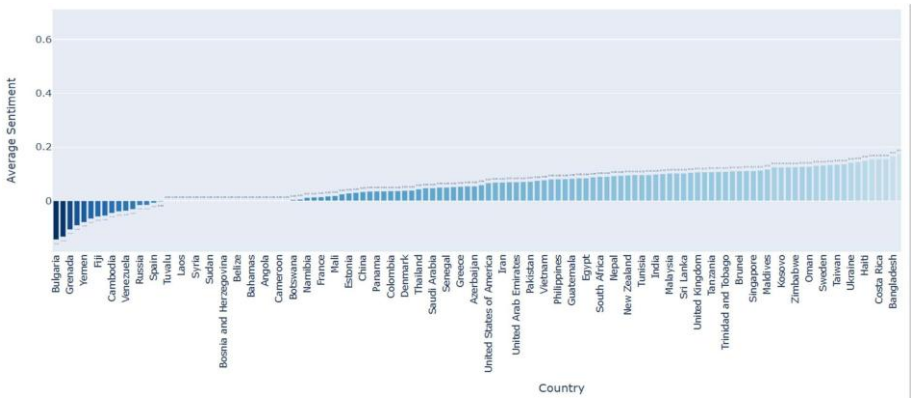


Figure 3. Sentiment score by User Locations

4.1.3 Hashtag Analysis of COVID-19 Vaccine Discourse

This analysis focuses on the most frequently used hashtags related to COVID-19 vaccines, offering insights into both global and regional conversations around the pandemic. By identifying these key hashtags, we can better understand the prevailing themes and concerns expressed in public discussions, ranging from worldwide

vaccination campaigns to localized issues involving specific vaccine types and distribution challenges.

Prominent hashtags such as #covidvaccine and #covid19 dominate the discourse, highlighting the global emphasis on vaccine distribution and pandemic control efforts. Vaccine-specific hashtags like #covishield and #covaxin point to the significant attention given to particular vaccine brands. The hashtag #vaccine captures broader conversations about vaccination in general. Regional hashtags, such as #bengaluru, underscore local concerns related to vaccine rollout logistics, while #coronavirus reflects ongoing global focus on the virus and its evolving variants. The diversity of hashtags reveals a rich mix of global and local narratives, emphasizing critical topics such as vaccine accessibility, safety, and governmental responses to the pandemic [18].

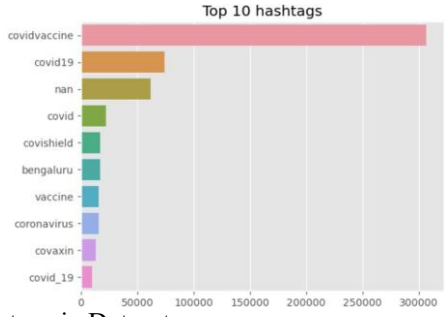
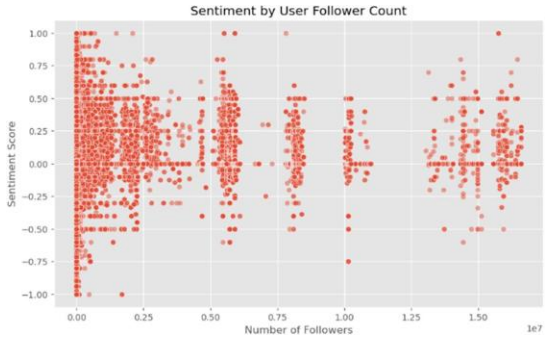


Figure 4. Top 10 Hashtags in Dataset

4.1.4 Relationship Between Sentiment Score and Number of Followers

This analysis investigates whether a user's follower count influences the sentiment expressed in their social media posts. By examining the relationship between follower count and sentiment scores, we aimed to determine if users with larger audiences tend to post with a particular tone. A scatter plot (Figure 5) visualizes sentiment scores across varying follower counts. The plot reveals a wide distribution of sentiment values—ranging from negative to positive—across all follower count ranges. Despite this spread, no clear correlation emerges between the number of followers and the sentiment expressed. These findings suggest that the sentiment of tweets is largely independent of user influence measured by follower count. Instead, sentiment appears to



be driven primarily by the content of the message and its context rather than the size of a user's audience.
Figure 5. Sentiment by User follower count

4.1.5 Tracking Vaccine Mentions on Twitter

This analysis examines the frequency of mentions of influential users in COVID-19-related social media discussions. The volume of mentions reflects the significant roles these users play in shaping public discourse and disseminating critical information about the pandemic and vaccine rollout. Figure 6 presents the top-mentioned users in COVID-19 vaccine tweets. Key accounts such as @CDCgov, @WHO, and @POTUS dominate due to their leadership in providing vital health information, guidelines, and policies throughout the pandemic. The CDC and WHO, as authoritative health organizations, guided vaccination strategies globally, while the U.S. President (POTUS) was central to national vaccine distribution and public communication. The frequent mentions of @Pfizer highlight its pivotal role as a leading vaccine developer, making it a key topic of discussion related to vaccine availability and efficacy. In India, @MoHFW_INDIA (Ministry of Health and Family Welfare) and @narendramodi (Prime Minister Narendra Modi) were commonly referenced, reflecting the government's active role in vaccine distribution and public health strategies closely followed by citizens. The prominence of @YouTube indicates the platform's widespread use for sharing vaccine-related content, including educational videos, tutorials, and official updates. Mentions of @JoeBiden, @US_FDA, and @BorisJohnson underscore the global focus on vaccine policies and decisions made by U.S. and U.K. leaders and regulatory bodies, which have a significant impact on public confidence and vaccine uptake. Overall, these mentions demonstrate a clear public reliance on authoritative sources for accurate information, policy guidance, and updates regarding the vaccine rollout process.

Top Users Mentioned:

mentions	
CDCgov	2420
WHO	2025
POTUS	2005
pfizer	1937
MoHFW_INDIA	1508
YouTube	1411
narendramodi	1407
JoeBiden	1199
US_FDA	1168
BorisJohnson	1168

Figure 6. Top Users Mentioned

4.1.6 Thematic Word Clouds and Analysis

Thematic word clouds were created to visualize and compare the most frequently mentioned terms in tweets from India and the USA, shedding light on regional narratives surrounding the COVID-19 vaccination discourse. The word clouds, shown in Figures 7 and 8, provide insights into the thematic content that shaped public discussions in both countries [17].

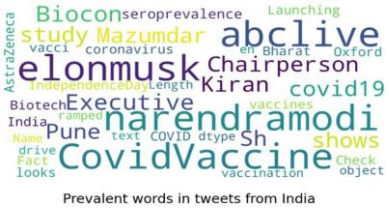


Figure 7. Prevalent words in tweets from India

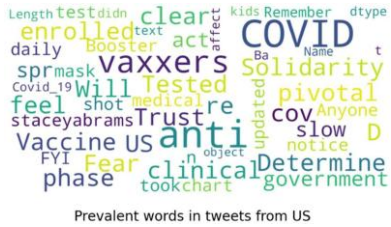


Figure 8. Prevalent words in tweets from the USA

In the Indian tweets, the term “abclive” appeared frequently. Upon further analysis, it was identified as a media outlet providing real-time pandemic updates, highlighting the media's significant role in shaping vaccine discourse. Names like “narendramodi”, referring to the Indian Prime Minister, and keywords such as “covidshield” were also prominently featured. The frequent mention of “narendramodi” reflects the strong public association of the vaccination campaign with political leadership, indicating that vaccine rollout in India was as much a political issue as a public health one. Public discourse often included expressions of both support and criticism of government efforts, with the Prime Minister seen as a key figure in these conversations. The keyword “covidshield” points to focused discussions around the Oxford-AstraZeneca vaccine manufactured in India, underscoring its significant role in the country’s immunization program. Conversely, tweets from the USA prominently featured terms like “cdcgov”, “vaccine”, and “publichealth”, signaling a stronger focus on institutional communication and public health messaging. This emphasizes the role of health agencies such as the CDC in guiding and shaping the vaccination narrative, reflecting public reliance on authoritative sources for vaccine information.

4.1.7 N-Grams

To better understand recurring word patterns and associations within the dataset, an N-gram analysis was conducted using unigrams (1-gram), bigrams (2-gram), and trigrams (3-gram). This linguistic approach helps identify frequently discussed topics and reveals relationships between terms, offering deeper insights into the public discourse surrounding COVID-19 vaccines.

Unigrams represent individual words. As shown in Figure 9, frequently occurring terms such as “vaccine”, “slots”, and “vaccinated” highlight the central focus of the discussion—vaccination itself, the process of getting vaccinated, and the availability of vaccine slots.

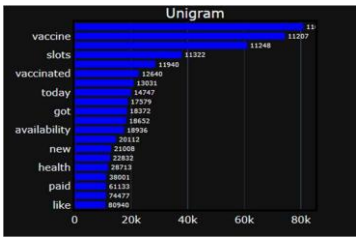


Figure 9. Unigram analysis

Bigrams capture pairs of consecutive words, providing context for how terms are used together. Figure 10 reveals common bigrams such as "covid vaccine", "paid slots", and "availability pmfree", which reflect public conversations around vaccine access, booking systems, and service models—particularly the distinction between free and paid options.

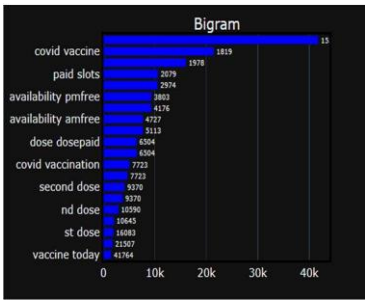


Figure 10. Bigram analysis

Trigrams consist of three-word sequences, offering even more specific insights into discourse structure. As illustrated in Figure 11, examples like "slots dose dose", "availability pmfree slots", and "dose dosepaid slots" suggest highly detailed conversations about appointment availability, vaccine dosage, and logistical concerns regarding the distribution process.

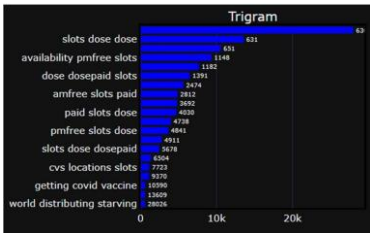


Figure 11. Trigram analysis

This N-gram analysis provides a granular view of how people framed their concerns, sought information, and communicated about vaccine logistics on social media platforms during the pandemic.

4.2 Sentiment Analysis

Sentiment analysis involves categorizing textual data into predefined sentiment

classes—typically positive, neutral, or negative. In this study, sentiment classification was carried out using a range of machine learning algorithms. As previously discussed in Section 3, the dataset exhibited significant class imbalance, which necessitated the application of balancing techniques such as Random Under Sampling, Random Over Sampling, and Synthetic Minority Over-sampling Technique (SMOTE).

Following data balancing, TF-IDF vectorization was employed to convert text into numerical features. Due to the high dimensionality of the resulting feature set, feature selection techniques were introduced to improve model efficiency and generalisability. Specifically, Variance Threshold and SelectKBest were used to reduce the number of features from over 15,000 to 2,500. This selection ensured that only the most relevant features informed the training process [20, 21].

Two supervised learning models were selected for comparative evaluation: Logistic Regression and Multinomial Naive Bayes. Hyperparameter optimisation was conducted using GridSearchCV, ensuring the best parameter configuration for each algorithm. For Logistic Regression, the optimal parameters were a regularisation strength (C) of 1.0 with L2 regularisation, balancing bias and variance effectively. For Multinomial Naive Bayes, a smoothing parameter (alpha) of 0.01 yielded optimal results, improving the model's ability to manage infrequent terms. The dataset was split into 80% training and 20% testing subsets. TF-IDF transformed the text data by assigning greater weight to discriminative terms, while common terms like "the" or "is" were de-emphasised. Post-vectorisation, Variance Threshold removed low-variance features, and SelectKBest retained the top 500 most informative features.

Model performance was assessed using standard evaluation metrics: accuracy, precision, recall, and F1-score. While accuracy provides a general performance overview, precision, recall, and F1-score offer deeper insights, especially important in imbalanced datasets. Confusion matrices were also generated to visualise classification errors.

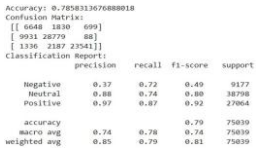


Figure 12: Logistic Regression Accuracy

Results: Logistic Regression (Figure 12)

The Logistic Regression model achieved an overall accuracy of 78.6%.

Positive sentiment: Precision 97%, Recall 87%, F1-score 92%

Neutral sentiment: Precision 88%, Recall 74%

Negative sentiment: Precision 37%, Recall 72%, F1-score 49%

While performance was high for positive sentiment, the model struggled to correctly classify negative instances, often confusing them with neutral ones.

```
Accuracy: 0.7304601607164275
Confusion Matrix:
 [[ 6947  1787   443]
 [ 9952 28362   484]
 [ 4707  2853 19504]]
Classification Report:
              precision    recall  f1-score   support

    Negative       0.32      0.76      0.45      9177
     Neutral       0.86      0.73      0.79     38798
    Positive       0.95      0.72      0.82     27064

    accuracy                           0.73     75039
   macro avg       0.71      0.74      0.69     75039
weighted avg       0.83      0.73      0.76     75039
```

Figure 13: MultinomialNB Accuracy
Results: Multinomial Naive Bayes (Figure 13)
The Multinomial Naive Bayes model achieved a slightly lower accuracy of 73.05%.
Positive sentiment: Precision 95%, Recall 72%, F1-score 82%
Neutral sentiment: Precision 86%, Recall 73%
Negative sentiment: Precision 32%, Recall 76%, F1-score 45%



Figure 14: Logistic Regression Confusion Matrix

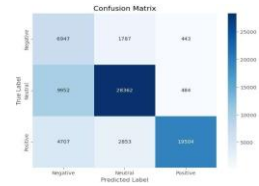


Figure 15: MultinomialNB Confusion Matrix

MultinomialNB was effective in classifying neutral sentiment but misclassified many positive sentiments as neutral. Both models exhibited difficulties distinguishing between negative and neutral sentiments, as shown in the respective confusion matrices (Figures 14 & 15). However, Logistic Regression demonstrated better overall balance and interpretability. In summary, this comparative analysis highlights the importance of addressing class imbalance, applying effective feature selection, and tuning model parameters. Logistic Regression emerged as the most suitable model, particularly for capturing positive sentiments with high reliability. However, improvements are needed in handling negative sentiment, suggesting future work should explore ensemble methods or deep learning approaches for enhanced performance [22].

4.3 Dynamic Sentiment Trends in Social Media

In the digital age, understanding public sentiment through social media analysis has

become essential for governments, organisations, and businesses seeking to monitor public opinion and respond to societal concerns. This section explores dynamic sentiment trends using social media data, focusing on how sentiment evolves over time in response to major events—specifically the COVID-19 pandemic.

To begin, raw social media text data was preprocessed to address issues such as missing values, noise, and invalid entries. Only clean, valid textual content was retained for sentiment analysis. The VADER (Valence Aware Dictionary and sEntiment Reasoner) tool was selected due to its effectiveness in handling social media text, particularly tweets. Each post was scored using VADER's compound polarity score and categorised into positive, neutral, or negative sentiment. These individual scores were then aggregated by date to construct a time series representation of sentiment trends. Figure 16 visualises these trends, showing the evolution of sentiment across key time points.

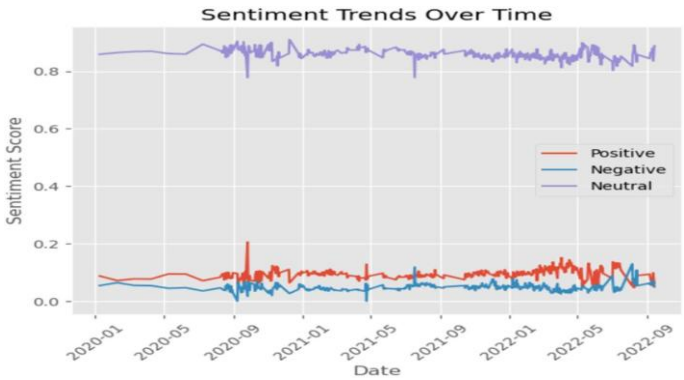


Figure 16: Sentiment trends over time

Key Observations from Sentiment Trends:

A marked spike in positive sentiment was observed in late 2020, coinciding with the rollout of COVID-19 vaccines. Social media posts during this period reflected feelings of hope, relief, and optimism. Simultaneously, a rise in negative sentiment occurred due to public concerns about vaccine side effects, inequities in access, and government handling. Discussions were often fueled by misinformation, safety fears, and logistical challenges.

Throughout the pandemic, public sentiment fluctuated in response to key milestones such as vaccine approvals, lockdown measures, and health announcements. For instance, positive sentiment peaked following initial vaccine approvals, while negative sentiment grew amid reports of side effects and policy debates. Recurring emotional themes—including fear, anxiety, and hope—dominated social media discourse. These emotions mirrored the real-world impact of the pandemic, as individuals navigated uncertainty, lockdowns, and eventual relief with the arrival of vaccines. By tracking COVID-19-related sentiment on social media, this analysis provided valuable insights into public mood and emotional response throughout the pandemic. Such real-time monitoring of sentiment can inform public health communication strategies, helping

authorities better engage with the public during crises. Furthermore, this methodology can be extended to other domains such as product feedback, election monitoring, or crisis response, making it a versatile tool for opinion mining.

5. Conclusion and Future Section

This study investigated the application of sentiment analysis techniques to track and analyze dynamic sentiment trends within social media discourse. The findings underscore the potential of machine learning (ML) and natural language processing (NLP) to capture public sentiment, particularly in real time via social media platforms. Key challenges—such as imbalanced datasets, feature engineering from unstructured text, and model selection—were addressed using methods like SMOTE for data balancing, TF-IDF for feature extraction, and feature selection techniques such as Variance Threshold and SelectKBest.

Among the models tested, **Logistic Regression** emerged as the most effective in classifying sentiment, especially for extreme cases (positive and negative), achieving an accuracy of **78.58%** and an F1-score of **92% for positive** and **49% for negative** sentiment. **Multinomial Naive Bayes**, while less effective for extreme sentiments, performed well for **neutral sentiment** detection with **73.05% accuracy**, highlighting the value of combining models to capture different sentiment categories. Hyperparameter tuning further contributed to improved model performance. The second part of the research explored **temporal sentiment trends**, revealing that sentiment on social media is highly responsive to external events, such as major news or viral topics. By mapping these fluctuations over time, we demonstrated how online sentiment shifts in response to real-world developments. This approach offers actionable insights for organizations, businesses, and policymakers seeking to adapt strategies, tailor messaging, or manage crises based on public mood. For example, marketing campaigns can be aligned with positive sentiment, while public health messaging can address emerging concerns.

Future Directions: Building on the results of this study, future work should explore:

Advanced ML/DL Models: Incorporating deep learning architectures, such as **Recurrent Neural Networks (RNNs)** and **transformer-based models like BERT**, can enhance sentiment classification by capturing deeper contextual and linguistic nuances, especially in longer or more complex texts [23]. **Causal Inference**: Future studies could apply **causal modelling techniques** to identify the factors driving sentiment shifts. This would help move beyond descriptive analysis to uncovering the underlying causes of public sentiment changes—such as demographic patterns, events, or policy announcements—enabling more targeted responses. **Broader Applications**: While this study focused on COVID-19-related tweets, the methodology is transferable to other public health domains. For example, sentiment analysis could be used to monitor **vaccine hesitancy**, evaluate **public health campaigns**, or detect **mental health trends** based on

online discussions. Real-time monitoring of sentiment can guide public communication strategies and help address emerging issues proactively. **Multimodal Sentiment Analysis**: As social media content diversifies beyond text to include **images, video, and audio**, sentiment analysis must evolve. **Multimodal models**, which integrate visual and auditory cues with textual data, could offer a more holistic understanding of sentiment, particularly on platforms like **Instagram, YouTube, and TikTok**. **Domain-Specific NLP Enhancements**: The inclusion of **emotion detection models**, **sentiment lexicons**, and domain-specific **NLP tools** can further refine the granularity and relevance of sentiment analysis, particularly in sectors like healthcare where nuanced emotional understanding is crucial.

In conclusion, this research demonstrates the value of combining ML and NLP techniques for dynamic sentiment analysis and lays the groundwork for future innovations that enhance the accuracy, applicability, and impact of such models in real-world scenarios.

References

1. Kanade, V. What is Sentiment Analysis? Definition, Tools, and Applications. *Spiceworks*, 2022. Available online: https://www.spiceworks.com/tech/artificial-intelligence/articles/what-is-sentiment-analysis/
2. Nasukawa, T.; Yi, J. Sentiment analysis. In *Proceedings of the International Conference on Knowledge Capture (K-CAP '03)*, Sanibel Island, FL, USA, 23–25 October 2003. https://doi.org/10.1145/945645.945658.
3. Pandey, A.R.; et al. Advanced Sentiment Analysis for Managing and Improving Patient Experience: Application for General Practitioner (GP) Classification in Northamptonshire. *Int. J. Environ. Res. Public Health* 2023, *20*(12), 6119. https://doi.org/10.3390/ijerph20126119. Available online: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC10298432/
4. AdmiralKhairul. Sentiment Analysis in Healthcare Organizations | Within3. *Within3*, 18 May 2022. Available online: https://within3.com/blog/sentiment-analysis-applications-healthcare
5. Xu, J.; et al. Citation Sentiment Analysis in Clinical Trial Papers. *AMIA Annu. Symp. Proc.* 2015, *2015*, 1334. Available online: https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4765697/
6. Ng QX, Lim SR, Yau CE, Liew TM. Examining the Prevailing Negative Sentiments Related to COVID-19 Vaccination: Unsuper- vised Deep Learning of Twitter Posts over a 16 Month Period. Vaccines (Basel). 2022 Sep 2;10(9):1457. doi: 10.3390/vac- cines10091457. PMID: 36146535; PMCID: PMC9503543.
7. Bulut O, Poth CN. Rapid assessment of communication consistency: sentiment analysis of public health briefings during the COVID-19 pandemic. AIMS Public Health. 2022 Feb 10;9(2):293-306. doi: 10.3934/publichealth.2022020. PMID: 35634025; PMCID: PMC9114780.
8. Ali GGMN, Rahman MM, Hossain MA, et al. Public Perceptions of COVID-19 Vaccines: Policy Implications from US Spatio- temporal Sentiment Analytics. Healthcare. 2021;9(9):1110. https://doi.org/10.3390/healthcare9091110

9. Bansal, A., Choudhry, A., Sharma, A., & Susan, S. (2024). Sentiment and word cloud analysis of tweets related to COVID-19 vaccines before, during, and after the second wave in India. https://www.worldscien- tific.com/doi/epdf/10.1142/9789811289125_0010
10. Cheng T, Han B and Liu Y (2023) Exploring public sentiment and vaccination uptake of COVID-19 vaccines in England: a spatiotemporal and sociodemographic analysis of Twitter data. *Front. Public Health*. 11:1193750. doi: 10.3389/fpubh.2023.1193750
11. Molenaar, A.; Lukose, D.; Brennan, L.; Jenkins, E.; McCaffrey, T. Using Natural Language Processing to Explore Social Media Opinions on Food Security: Sentiment Analysis and Topic Modeling Study. *J. Med. Internet Res.* 2024, 26, e47826. https://doi.org/10.2196/47826.
12. Thakur, N., S. Cui, V. Knieling, K. Khanna, and M. Shao. "Investigation of the Misinformation about COVID-19 on YouTube Using Topic Modeling, Sentiment Analysis, and Language Analysis." *Computation* 12, no. 2 (2024): 28. https://doi.org/10.3390/computation12020028.
13. Alshanik, F.; Khasawneh, R.; Dalky, A.; Qawasmeh, E. Unveiling Topics and Emotions in Arabic Tweets Surrounding the COVID-19 Pandemic: Topic Modeling and Sentiment Analysis Approach. *JMIR Infodemiology* 2025, 5, e53434. https://doi.org/10.2196/53434.
14. Parveen, S.; Pereira, A.; Garzon-Orjuela, N.; McHugh, P.; Surendran, A.; Vornhagen, H.; Vellinga, A. COVID-19 Public Health Communication on X (Formerly Twitter): Cross-Sectional Study of Message Type, Sentiment, and Source. *JMIR Form. Res.* 2025, 9, e59687. https://doi.org/10.2196/59687.
15. kaushiksuresh147. COVID Vaccine Tweets. Available online: https://www.kaggle.com/datasets/kaushiksuresh147/covidvac- cine-tweets (accessed on 22 February 2024).
16. Papia, S.K.; Khan, M.A.; Habib, T.; Rahman, M.; Islam, M.N. DistilRoBiLSTMFuse: An Efficient Hybrid Deep Learning Ap- proach for Sentiment Analysis. *PeerJ Comput. Sci.* 2024, 10, e2349. https://doi.org/10.7717/peerj-cs.2349.
17. Ahmed S, Khan DM, Sadiq S, Umer M, Shahzad F, Mahmood K, Mohsen H, Ashraf I. Temporal analysis and opinion dynamics of COVID-19 vaccination tweets using diverse feature engineering techniques. PeerJ Comput Sci. 2023 Mar 10;9:e1190. doi: 10.7717/peerj-cs.1190. PMID: 37346678; PMCID: PMC10280254.
18. Fuster-Casanovas A, Das R, Vidal-Alaball J, Lopez Segui F, Ahmed W. The #VaccinesWork Hashtag on Twitter in the Context of the COVID-19 Pandemic: Network Analysis. JMIR Public Health Surveill. 2022 Oct 28;8(10):e38153. doi: 10.2196/38153. PMID: 36219832; PMCID: PMC9620955.
19. Vishwakarma A, Chugh M. COVID-19 vaccination perception and outcome: society sentiment analysis on twitter data in India. Soc Netw Anal Min. 2023;13(1):84. doi: 10.1007/s13278-023-01088-7. Epub 2023 May 10. PMID: 37193096; PMCID: PMC10170045.
20. Park B, Jang IS, Kwak D. Sentiment analysis of the COVID-19 vaccine perception. Health Informatics J. 2024 Jan- Mar;30(1):14604582241236131. doi: 10.1177/14604582241236131. PMID: 38403926.
21. Melton CA, White BM, Davis RL, Bednarczyk RA, Shaban-Nejad A. Fine-tuned Sentiment Analysis of COVID-19 Vaccine-Re- lated Social Media Data: Comparative Study. J Med Internet Res. 2022 Oct 17;24(10):e40408. doi: 10.2196/40408. PMID: 36174192; PMCID: PMC9578521.
22. Ansari MTJ, Khan NA. Worldwide COVID-19 Vaccines Sentiment Analysis Through Twitter Content. Electron J Gen Med. 2021;18(6):em329. https://doi.org/10.29333/ejgm/11316

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